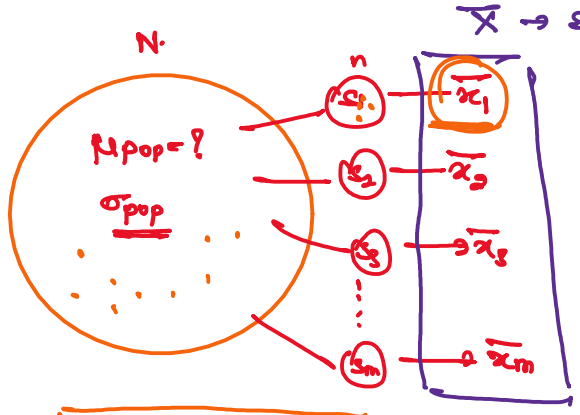
Wednesday, 24 December 2025 6:04 PM

- Inferential statistics.
  - we have discussed how to find the estimation
    - Point Estimate.
      - Point estimate with a single sample.
      - Point estimate with m sample using Central limit theorem (CLT).
    - Confidence Interval estimate.
      - Confidence interval estimate with a single sample with CLT → it work if and only if either population variance or population standard deviation is known.

$\bar{X}$ → sampling distribution of sample mean.

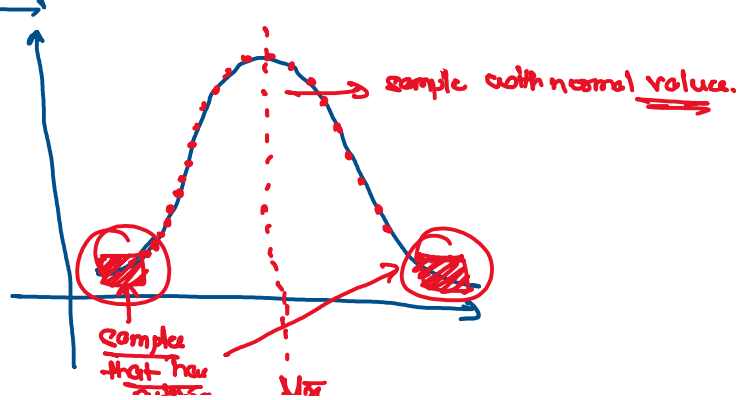


1. $\boxed{\mu_{population} \approx \bar{x}_1}$ → Bias estimate.

2. According to CLT.
   - $\bar{X} \sim N(\mu_{\bar{x}}, \sigma_{\bar{x}})$.
   - $\mu_{\bar{x}} \approx \mu_{population}$.
   - $\sigma_{\bar{x}} \approx \frac{\sigma_{population}}{\sqrt{n}}$. → sample size.

$\boxed{\mu_{population} \approx \mu_{\bar{x}}}$ → Data collection is hence

uncertainity.



mean of any sample

$\bar{x}_i$ is a part of the distribution → $\bar{X}$ is a distribution of all the $\bar{x}_i$

$$\boxed{\mu_{population} \approx \bar{x}_i \pm \text{margin.}}$$

with y% confidence Interval.

- 68% confidence → $\mu_{pop} \approx \bar{x}_i \pm 1\sigma_{\bar{x}}$
- 95% confidence → $\mu_{pop} \approx \bar{x}_i \pm 2\sigma_{\bar{x}}$
- 99.7% confidence → $\mu_{pop} \approx \bar{x}_i \pm 3\sigma_{\bar{x}}$

In general term

$$\boxed{\mu_{pop} \approx \bar{x}_i \pm \underbrace{z \times \sigma_{\bar{x}}}_{\text{margin of error}}}$$

- According to CLT $\sigma_{\bar{x}} \approx \boxed{\dfrac{\sigma_{population}}{\sqrt{n}}}$ → standard error

replace $\sigma_{\bar{x}}$ with $\dfrac{\sigma_{population}}{\sqrt{n}}$

$$\boxed{\mu_{population} \approx \bar{x}_i \pm z \times \frac{\sigma_{population}}{\sqrt{n}}}$$

here you need to the population standard deviation

- Margin of error v/s Confidence Interval.
  - Margin of error → maximum expected estimated error from the sample statistic.
  - It tell us how far the estimate can deviate from the true population parameter.
  - Confidence Interval → it is a range of values within which the population parameter is expected to lie.
  - It tell us where the true population parameter is likely to be

C.I = point estimate ± margin of error

MOE = Critical value × standard error.

Critical value → z-value / t-value

- Relationship b/w Margin of error & Confidence Interval.
  - Mathematical → C.I → margin

Lower bound < mean − margin of error
upper bound < mean + margin of error.

- Margin of error determines the width of the confidence interval.

- As we increase or decrease the margin of error, we increase or decrease Z-value

$$M.OE = Z \times \frac{\sigma_{pop}}{\sqrt{n}}$$ → constant term.

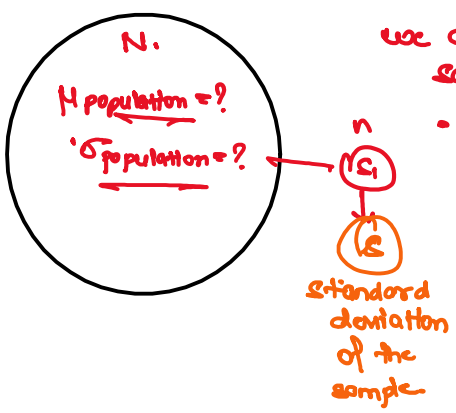
68% → Z = 1
95% → Z = 2
99.7% → Z = 2.

Z — critical value (x-state / Z-score.)

- As we increase Z-value or decrease Z-value → confidence interval also increase or decreased.

90% confidence interval — Z critical value → no we have refer a x-table.

- What if $\sigma_{population}$ is not known / unknown ?.



we can replace $\sigma_{population}$ with sample standard deviation.

- as we replace σ_{pop} to s standard. bias will increase, to cater that or cancel that we also have to increase the margin of error.
- To increase the margin of error we will replace Z-critical value with t-critical value.

So, when the population standard deviation is not known.

$$\boxed{\mu_{pop} \approx \bar{x}_s \pm t \times \frac{s}{\sqrt{n}}}$$

s → sample standard deviation
n → sample size

CDF → Cumulative Distribution function

value → output = probability.

PPF → Percent point function. → inverse of CDF

probability → value (output).

→ In C.I., we already know the probability (eg. 0.975 for 95% CI)
→ x-value
→ probability

we want a x-value corresponding to that probability.

ppf.

68% → x=1
95% → x=2.
99.7% = x=3

significance level = 1 − confidence interval
1 − 0.95
= 0.05.

x-score = stats.norm.ppf (1 − significance level/2). → two tail test
(0.975).

for one tail test
1 − significance level.